

# Design and Analysis of Algorithms

## Milestone 1: Project Idea Submission

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### Selected Topic: Maximum Subarray Sum Problem

The Maximum Subarray Sum problem involves finding a contiguous subarray within a given array of integers (which may include both positive and negative values) such that the sum of its elements is maximized. The objective is to determine the subarray that produces the largest possible sum among all possible contiguous subarrays.

This problem is a fundamental concept in algorithm design and analysis, as it demonstrates how different algorithmic approaches can significantly improve efficiency, from brute force methods to optimized dynamic programming techniques. It has practical applications in various fields, including financial analysis (maximum profit calculation), signal processing, bioinformatics, and image processing.

### Algorithms to Be Compared

Our group will implement and compare the following five algorithms, each representing a distinct design paradigm:

| # | Algorithm                        | Design Technique    | Time Complexity |
|---|----------------------------------|---------------------|-----------------|
| 1 | Brute Force (All Sub arrays)     | Exhaustive Search   | $O(n^3)$        |
| 2 | Better Brute Force (Running Sum) | Exhaustive Search   | $O(n^2)$        |
| 3 | Divide & Conquer                 | Divide & Conquer    | $O(n \log n)$   |
| 4 | Kadane's Algorithm               | Dynamic Programming | $O(n)$          |
| 5 | 2D Maximum Subarray              | DP + Extension      | $O(n^3)$        |

In this project, multiple algorithms with varying time complexities will be implemented and compared to analyze their efficiency and performance.